# Importance of N2Hx Reaction Pathways in Modelling of NH3/H2/air Flame Chemistry

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**ABSTRACT**

To investigate the importance of N2Hx reaction pathways in modelling NH3/H2/air flames, the $S_L$ of the mixture was modelled using an existing reaction mechanism. This mechanism reproduced $S_L$ of NH3/air and NH3/CH4/air flames accurately as the mechanism was optimized and validated for these fuel blends however, $S_L$ of NH3/H2/air were under-predicted. Comparing reaction pathways and $S_L$ sensitivities of other mechanisms that were optimized for NH3/H2/air, the importance of N2Hx reaction pathways was established. A present mechanism was developed and then validated with data from literature.

A reduced chemical kinetics mechanism capable of modeling the unstretched laminar burning velocity of ammonia and its blends was developed. Existing mechanisms were studied using reaction pathway analysis and normalized sensitiviy of burning velocity to rate constants of reactions. The developed mechanism utilized a detailed N2Hx chemistry to model

## 1. Introduction

(1) Ammonia has gained interest as a carbon free fuel for combustion systems and the blending of ammonia with other faster burning fuel, such as methane and hydrogen, is a method employed to improve the combustion characteristics of ammonia. To further study the phenomena observed with the combustion of ammonia and its blends under varying conditions, numerical simulations with chemical kinetics mechanisms is neccessary. The choice of mechanism determines the accuracy of the numerically simulated flames. Reaction mechanisms that are able to accurately predict parameters of the flame, such as laminar burning velocity ($S_L$), under varying conditions, like rich/lean equivalence ratios and varying fuel compositions, is desired.

(*1) Ammonia has gained interest as a carbon free fuel for combustion systems and the blending of ammonia with other faster burning fuel, such as methane and hydrogen, is a method employed to improve the combustion characteristics of ammonia. In order to develop practical applications with these fuels, numerical simulations that can accurately evaluate the unstretched laminar burning velocity of the flame is required as this flame property can be used to describe various premixed flame phenomena. Consequently, the development of chemical kinetics mechanisms that can accurately model this property at varied initial conditions are required. Moreover, reaction mechanisms with fewer species and reactions are desirable as large-scale combustion models and computational fluid dynamics simulations that employ bigger chemical kinetics mechanisms involve significantly more computational cost.

(*2) Numerous chemical kinetic mechanisms that model NH3/air flames or specific blends of ammonia exist. However, most of these mechanisms were developed, optimized and validated for targeted mixtures, specific set of flame characteristics and select experimental conditions. For instance, a reduced mechanism developed by Okafor et al. [1], a chemical kinetics mechanism derived from a detailed mechanism by Okafor et al. [2] and was optimized for NH3/CH4/air flames. This mechanism satisfactorily predicts $S_L$ at equivalence ratios ranging from lean to rich for NH3/air flames [1, 3] and NH3/CH4/air flames [1, 4]. This same reaction mechanism, however, is found to over-predict ignition delay times for NH3/air combustion [5] and $S_L$ of NH3/H2/air flames for hydrogen concentrations greater than $X_{H2} \approx 0.4$ where $X_{H2}$ is the molar fraction of the specie $H_2$ in the fuel. The detailed mechanism by Gotama et al. [6], a mechanism based on Han et al.'s mechanism [7], was optimized for NH3/H2/air combustion and then was validated for xH2$\approx$ 0.4 at rich, stoichiometric and lean equivalence ratios [6]. Moreover, the mechanisms displayed accurate $S_L$ predictions for H2/air flames at various equivalence ratios at atmospheric pressure. Gotama et al. also developed a reduced mechanism (26 species and 119 reactions) based on their detailed mechanism [6]. The mechanism by Nakamura et al. [8] was developed with Miller and Bowman's mechanism [9] as a base to which N2Hx chemistry was added. This led to accurate predictions NH3, NO and N2O profiles in NH3/air flames. NH3/air flame speed was reasonably predicted by this mechanism at lean and stoichiometric conditions [3] too. Another study found Nakamura Mech's $S_L$ predictions of NH3/H2/air flames were satisfactory too, especially in fuel-rich conditions ($\phi$ =1.1–1.4). However, this mechanism struggles to accurately model NO and N2O profiles in the reaction zones of NH3/air flames. Hence, a demand for a reduced reaction mechanism that can accurately evaluate the unstretched laminar burning velocity of various ammonia blends for a wide range of initial conditions exists.

(*3) This study aims to develop a reaction mechanism that can accurately model the $S_L$ of NH3/air, NH3/CH4/air and NH3/H2/air flames. The $S_L$, being a flame characteristics that dictates fuel compatability for a wide range of practical applications, is prioritized for this present mechanism. This mechanism should maintain accurate predictions for rich and lean equivalence ratios, elevated pressures and varied fuel compositions.

## 2. Numerical Procedure

(*1) In this study, the one-dimensional simulations for the unstretched laminar burning velocity, $S_L$, of premixed adiabatic flames were simulated using the Premixed Laminar Flame Speed Calculation Model in ANSYS Chemkin-PRO 2019 R3 and ANSYS Chemkin-PRO 2025 R2. The

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computation domain was set 10 cm and $S_L$ was calculated at the cold boundary of the computation domain. The adaptive grid setting GRAD and CURV were both set to 0.01. The ratio of H2 in the binary fuel, $X_{H2}$, is defined in 1. $X_{H2}$ was varied to study how each reaction mechanism's $S_L$ predictions are affected by hydrogen addition.

$$\mathrm{X_{H2}} = \frac{[\mathrm{H_2}]}{[\mathrm{NH_3}] + [\mathrm{H_2}]} \tag{1}$$

(*2)The software was also used to compute rate of production of species, species concentrations and sensitiviy of $S_L$ to rate constants of elementary reactions. The reduced chemical kinetics mechanism by Okafor et al. [1], Gotama et al.'s [6] detailed mechanism and Nakamura et al.'s [8] mechanism were used in the simulations. Henceforth, for the purpose of discussion, these mechanisms shall be referred to as Okafor Reduced Mech, Gotama Mech and Nakamura Mech respectively. A reduced chemical kinetic model based on Okafor Reduced Mech. [1] with 43 species and 152 reactions was developed as described in the following section. The developed mechanism will be refered to as Present Mech in this paper.

## 3. Development of the Present Mech

(*1) The new reaction mechanism developed in this paper was based on the Okafor Reduced mechanism. Okafor Reduced Mech was selected because this mechanism has been optimized and validated for $S_L$ prediction of NH3/air and NH3/CH4/air flames. This is true for varying equivalence ratios, pressures and, in the case of NH3/CH4/air flames, varying methane concentrations. This mechanism also models the $S_L$ of H2/air flames satisfactorily for most of these conditions. This mechanism performs poorly when the it is used to predict $S_L$ of NH3/H2/air flames at hydrogen concentrations greater than $X_{H2} = 0.3$ as depicted in Fig. 1, a comparison of numerically simulated values and experimental measurements of burning velociy at varying hydrogen concentrations. $X_{H2}$ on the horizontal axis is the molar fraction of the specie $H_2$ in the fuel. The mechanism's failure to predict the $S_L$ for this specific blend, despite accurately reproducing the individual components of NH3/H2/air flames, motivated the current investigation.

[Figure: lbv NH3/H2]

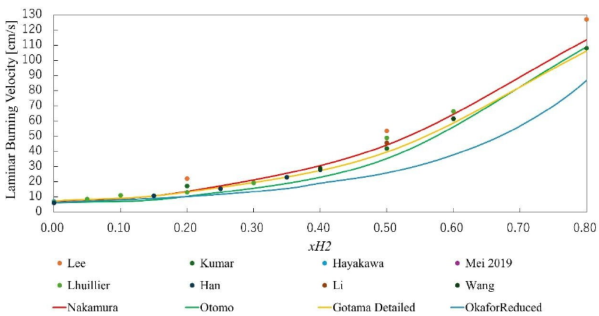


Fig. 1 Experimental and numerical laminar burning velocity of NH3/H2/air flames under stoichiometric condition at various hydrogen concentrations

(*2) Figure 1's comparison also highlights Gotama Mech and Nakamura Mech's $S_L$ predictions agree with experimental measurements of $S_L$ at all $X_{H2}$. Understanding where Gotama Mech and Nakamura Mech differ from Okafor Reduced Mech should provide insights as to where the under-prediction of NH3/H2/air flames stems from. In addition, development of Gotama Mech focused on improving NH3/H2/air flame speed prediction at intermediate hydrogen concentration [6]. Hence, the differences in how these mechanisms

(*3) To understand how oxidation kinetics of each fuel in the binary fuel interact with each other, reaction pathways for each mechanism was constructed using integrated rate of production (IROP) of major species related to NH3 and H2 oxidation. Each IROP is calculated by integrating the rate of production, $\dot{\omega}_{R,i}$, of specie $i$ in elementary reaction $R$ along the mechanism along the computational domin using Equation 2. The IROP, hence, characterized the contribution of each elementary reaction to the combustion of NH3/H2/air fuel in each chemical kinetics model.

$$IROP = \int_0^L \dot{\omega}_{R,i}\mathrm{d}x \tag{2}$$

(*4) Figures **??** are reaction pathways of stoichiometric NH3/H2/air flame of $X_{H2} = 0.5$ computed using Okafor Reduced Mech and Gotama Mech respectively. The thickness of each arrow corresponds to the relative thickness of each path's IROP to the path with the largest IROP, OH→H2O. Species besides each arrow have numbers that represent the percent contribution each elementary reaction has to the corresponding path. The pathways highlighted the key differences in how the interaction of NH3 and H2 oxidation paths are modelled in each mechanism to inform the development of Present Mech.

## 4. Results and Discussion

[Figures reaction pathways]

(*1) Comparisons between the pathways of Okafor Reduced Mech and Gotama Mech, Fig. **??** respectively, highlight interactions between NH3 and H2 oxidation pathways at intermediate H2 concentrations in the binary fuel are modelled differently in the two chemical kinetics models. For instance, reactions in the reaction path NH3↔N2H3 is modelled in Gotama Mech only. The forward reaction, the formation of N2H3, is given by;

$$\mathrm{NH_2 + NH = N_2H_3} \tag{R1}$$

and the reverse,the consumption of N2H3, is given by;

$$\mathrm{2NH_2 = N_2H_3 + H} \tag{R2}$$

. Gotama Mech's reaction pathway depicts the involvement of N2H3 in the control of intermediates important in ammonia oxidation pathway.

(*2) The reaction pathway NNH→N2 were distinct between Okafor Reduced Mech and Gotama Mech. Figure 2 shows that Gotama Mech exhibits an IROP approximately 2.2 times larger Okafor Reduced Mech and the pathway is primarily contributed to by the reaction;

$$\mathrm{NNH = N_2 + H} \tag{R3}$$

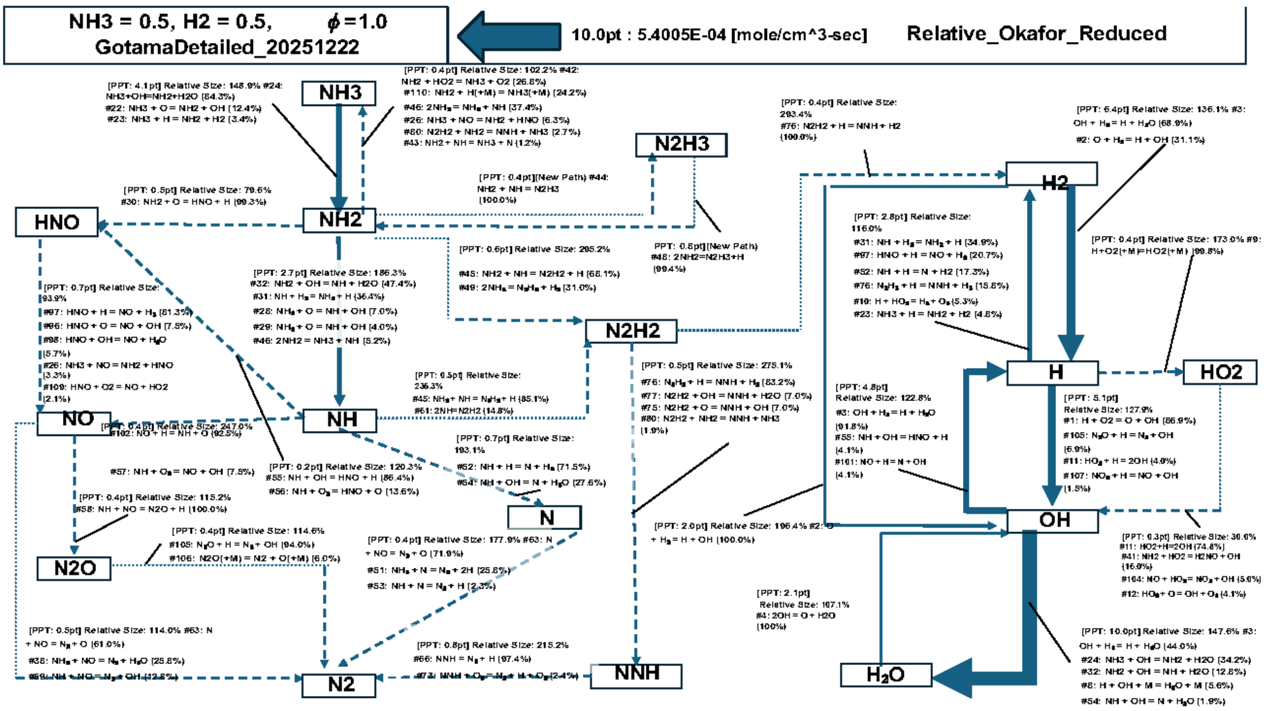


Fig. 2 Reaction pathway of NH3/H2/air flame at xH2=0.5 and equivalence ratio of 1.0 using Gotama Detailed Mechanism

in Gotama Mech. In Gotama Mech this pathway has a become significant contributor to H radical generation at intermediate hydrogen concentration.

(*3) Another reaction pathway, NH2→N2H2 can also be depicted to exhibit increased IROP in Gotama Mech, approximately 3 times larger than in Okafor Reduced Mech. The reaction;

$$NH_2 + NH = N_2H_2 + H \qquad \text{(R4)}$$

in the biggest contributor to the pathway. R4 is a H radical generating reaction while consuming NH2 and NH intermediates. Moreover, the increased N2H2 production further contributes to promote the pathway N2H2→NNH→N2 as the IROP for these pathways in Gotama Mech were approximately 2.8 times and 2.2 times that of Okafor Reduced Mech. In additon, (R1) was the significant reaction in the latter pathway suggesting .

(*5) N2Hx species' involvement in NH3 and H2 oxidation pathway interactions as hydrogen concentrations in an NH3/H2/air flame at $X_{H2} = 0.5$ was evident in Gotama Mech. Moreover, the general increase in IROP of N2Hx related reactions, relative to Okafor Reduced Mech's reaction pathway, along with the reactions and reaction pathways discussed in this section, it can be inferred that the N2Hx chemistry have been modelled very differently in the two chemical kinetics models. Hence, to reproduce this phenomena accurately, Present Mech was developed by adding the N2Hx chemistry from Gotama Mech [6] to Okafor Reduced Mech [1]. The reaction pathway of the Present Mech, hence, behaves similar to Gotama Mech when xH2=0.5 in the binary fuel. Figure 1 shows that, consequently, the Present Mech's $S_L$ predictions improve due to this and can now emulate experimental measurements of stoichiometric NH3/H2/air burning velocity accurately across all hydrogen concentrations at atmospheric pressure.

[Figure lbv sens analysis]

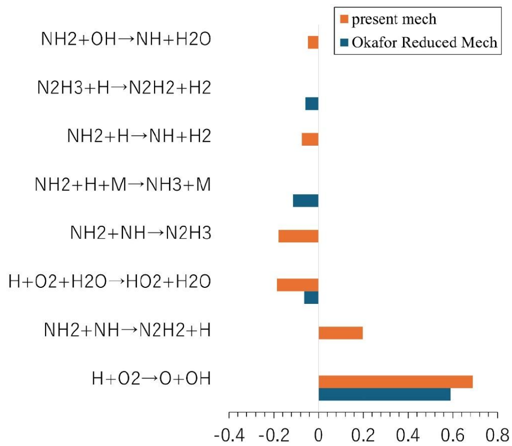


Fig. 3 sensitivity coefficients for laminar burning velocity of NH3/H2/air flames at xH2=0.5 and equivalence ratio of 1.0 under atmospheric conditions.

(*6) Figure 3 shows the normalized sensitivity of $S_L$ to rate constants of the elementary reactions using the Present Mech. Reaction (R4) is shown to play a significant role in promoting $S_L$ in Present Mech while remaining insignificant in Okafor Reduced Mech. In addition, reaction (R1) only significantly inhibits $S_L$ in the Present Mech while remaining insignificant Okafor Reduced Mech's sensitiviy analysis. The figure also shows other NHx and N2Hx related reactions being more significant with respect to $S_L$ in Present Mech at intermediate hydrogen concentration.

(*7) The reactions (R2), (R3) and (R4) involve N2Hx species and control H radical concentration — an important radical for promoting $S_L$. The increased IROP of these reactions in Present Mech imply an increase in H radical concentration which, correlates with the increased burning veloctiy predictions of this mechanism relative to Okafor Reduced Mech. Moreover, the formation and consumption of N2H3 with (R1) and (R2) respectively control NH and NH2 concentrations. NHx species are important as they interact with H and OH radicals and also produce these radicals. (R4) is an example of NHx species' involved in the production of an H radical. The other product of (R4), N2H2, is part of the pathway to form NNH which also produces H radicals via (R3). Moreover, reaction (R4)'s significant positive effect on the flame's burning velocity further indicates the importance of N2Hx species and their reaction pathways for fuel compostions of intermediate hydrogen concentration. Therefore, the Present Mech with a more accurate N2Hx chemistry is able to predict $S_L$ undere these conditions accurately

## 5. Validation

The Present Mech was validated with experimental data for $S_L$ of NH3/H2/air from the datasets consolidated by Han et al. [13]. These data sets were proven to be consistent with each other and widely used chemical kinetic mechanisms. Present Mech was validated for NH3/CH4/air flames also. Experimental data for NH3/CH4/air $S_L$ was compiled from from three datasets: Okafor et al. (2019) [1], Han et al. (2019) [14] and Ling et al (2026) [15]. However, experimental datasets, for both fuel blends, lacked ample $S_L$ measurements at elevated pressures, making validation of Present Mech for high pressure combustion difficult.

The results displayed accurate modelling of NH3/H2/air $S_L$ at all the equivalence ratios tested for all hydrogen concentration at atmospheric pressure. At 3 atm of pressure, intermediate H2 concentration (xH2=0.4, 0.6) were validated while, at 5 atm of pressure, only xH2=0.4 could be validated. This suggests that Present Mech is now capable of modelling flame speed of hydrogen enriched ammonia combustion. When comparing NH3/CH4/air predictions to measured $S_L$, the $S_L$ predictions are satisfactory. The model is able to match $S_L$ predictions to Okafor et al. (2019) [1] for all the its data points across elevated pressures, increasing xCH4 and different equivalence ratios. This implies that the underlying Okafor Reduced Mech optimizations for NH3/CH4/air flame chemistry in Present Mech is unaffected.

## 6. Concluding Remarks

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